Lesson plan · for the teacher

Pictures that fool a model

AI security · Level 3 · Lesson 3

Time
40 min

Materials
A stack of simple drawings on cards (circles, crosses, lines), or paper to draw them; A secret sorting rule written on a folded card; Board and chalk

Learning goals

- Explain how small, chosen changes fool a model.
- Test whether an answer is fragile with small nudges.
- Decide where a person must check a model's answer.

Activity: Fool the sorter

1. Warm-up (5 min): Ask pupils how they tell a cow from a goat. Then ask: what if you only ever saw cows on grass? Introduce the idea of a shortcut.
2. Teach (8 min): Explain that AI models also learn shortcuts, and that a small, chosen change can flip their answer while people see no difference. Give one example: stickers on a road sign.
3. Fool the sorter (15 min): A pupil plays the “model” with the secret rule card and sorts the class's drawings. Groups hunt for the smallest change that flips its answer without changing what the drawing looks like to people.
4. Toughen it (7 min): Reveal the rule. Groups rewrite it to be harder to fool (for example, look at the whole shape, not one corner) and test it again.
5. Wrap-up (5 min): Ask where a person should always check a machine's answer: marks, health, money, safety. Then give the exit ticket.

Make the secret rule depend on a tiny detail (a dot in one corner, a line slightly too long), just as models latch onto small details. Keep the drawings simple so every group can play.

Discussion

- Why could a tiny change fool our “model” but not us?
- If a model is 99% sure, should we trust it? When?
- Where should a person always have the final word?

Exit ticket

- Write one place where an AI's answer should never decide alone, and who should check it.

Try it at school tomorrow
Play “Fool the sorter”: one pupil sorts drawings by a secret rule, and the class hunts for the smallest change that flips the answer. Then talk about how to make the rule harder to fool, and why a person should check answers that matter.

Practice questions · for you

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1 **A school plans to take attendance with a face-recognition camera and nothing else. What is the safest plan?**

- ◯ Turn its confidence setting up
- ◯ Trust the camera: it's very accurate
- ◯ Use it as a helper, with a person checking and another way to sign in

2 **label(score) says "sick" from 0.5 up. Finish fragile(score, eps): it is True when nudging the score down by eps and up by eps gives two different labels.**

```
def label(score):
    if score >= 0.5:
        return "sick"
    return "healthy"

def fragile(score, eps):
    # True if a nudge of eps either way
    # changes the label
    return False

print(fragile(0.52, 0.05))
print(fragile(0.90, 0.05))
```

Write your program:

3 **Complete the program so it counts how many of the model's answers are fragile.**

```
scores = [0.91, 0.52, 0.08, 0.47, 0.66]
flips = 0
for s in scores:
    if _____(s, 0.05):
        flips = flips + 1
print(flips, "of", _____(scores))
```

Word bank: `len` `max` `fragile` `label`

4 **Which of these are adversarial examples: changed on purpose to fool a model? Choose all that apply.**

- ☐ Stickers placed on a road sign so a sign reader misreads it
- ☐ A sound people can hardly hear that steers a voice assistant
- ☐ A few letters swapped so a scam slips past a filter
- ☐ A blurry photo taken in the rain

5 **After adversarial training, a model can no longer be fooled.**

- ◯ True
- ◯ False

✎ **Where at your school, or in your pupils' lives, does an AI's answer decide something? What would happen if it could be fooled, and who checks it?**

Check your answers

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1 **Use it as a helper, with a person checking and another way to sign in**

Even an accurate model can be fooled or simply wrong about some pupils.

2

```
def label(score):
    if score >= 0.5:
        return "sick"
    return "healthy"

def fragile(score, eps):
    a = label(score - eps)
    b = label(score + eps)
    return a != b

print(fragile(0.52, 0.05))
print(fragile(0.90, 0.05))
```

If the labels a little below and a little above differ, the answer sits on a knife edge: an attacker, or plain noise, can flip it.

3

```
scores = [0.91, 0.52, 0.08, 0.47, 0.66]
flips = 0
for s in scores:
    if fragile(s, 0.05):
        flips = flips + 1
print(flips, "of", len(scores))
```

fragile(s, 0.05) tests each score; len(scores) is how many there are.

4 **Stickers placed on a road sign so a sign reader misreads it; A few letters swapped so a scam slips past a filter; A sound people can hardly hear that steers a voice assistant**

Stickers, swapped letters and hidden sounds are chosen on purpose to fool a model.

5 **False**

Adversarial training makes a model harder to fool (in the lab, about 14 of 18 attacks fall to about 7 of 18), but never impossible.